Alternating Vowel Paths

You are given an integer n. Count the number of valid strings of length n that consist only of lowercase English vowels ('a', 'e', 'i', 'o', 'u').

A string is valid if it satisfies both of the following conditions:

• It follows these directed transition rules from one character to the next:

o 'a' can only be followed by 'e'.

o 'e' can only be followed by 'a' or 'i'.

o 'i' can only be followed by 'a', 'e', 'o', or 'u'.

o 'o' can only be followed by 'i' or 'u'.

o 'u' can only be followed by 'a'.

• The transitions must strictly alternate in lexicographical direction. If a transition moves to a lexicographically greater vowel (for example, 'a' to 'e'), the immediately subsequent transition (if one exists) must move to a lexicographically smaller vowel (for example, 'e' to 'a'), and vice versa. Two consecutive transitions in the same direction are strictly forbidden.

Find the number of valid strings. Because the answer can be extremely large, return the valid count modulo 10^9+7.

Input Format

The first line contains a integer, n, denoting the length of the string.

Constraints

1 <= n <= 10^5

Output Format

The output prints an integer value

Sample Input 0

```
1
```

Sample Output 0

```
5
```

Explanation 0

The length is 1. Any single vowel is a valid string, and there are no transitions to evaluate. Thus, the valid strings are a, e, i, o, and u, totaling 5.

Sample Input 1

```
2
```

Sample Output 1

```
10
```

Explanation 1

The length is 2. The standard valid transitions are a to e, e to a, e to i, i to a, i to e, i to o, i to u, o to i, o to u, and u to a. Since there is only one transition in a string of length 2, the alternating direction rule does not restrict anything yet. There are exactly 10 valid strings.

Sample Input 2

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3
```

Sample Output 2

```
12
```

Explanation 2

The length is 3. We must apply the alternating direction rule. If a string starts with a to e, this is a move to a lexicographically greater vowel. From e, the only smaller valid transition is to a, forming aea. Moving e to i is invalid here because it would be two greater transitions in a row. By filtering all length 2 paths this way, exactly 12 valid strings remain.